Redbird Flight Simulations
Qualification and Approval Guide (QAG)

C510 Advanced Aviation Training Device
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Signature

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Section 1: Device Overview

The Redbird C510 is a versatile and affordable device that was designed to represent a Cessna Citation Mustang (CE-510). It is equipped with the following notable features:

- 3-axis electric motion platform providing pitch, roll and yaw motions.
- Wrap-around exterior visuals provided by 6 LCD screens
- Dual pilot controls including a 2-axis control-loaded yoke and interconnected rudder pedals
- Realistic switches, buttons, knobs, circuit breakers and other cockpit controls that are designed to match the aircraft wherever possible
- An interchangeable instrument panel to allow development of future configurations
- Closed Circuit intercom system, allowing for communication between the pilot, co-pilot and instructors using standard aviation headsets
- A portable instructors station, allowing the instructor to operate from inside or outside the simulator
- Optional supplemental oxygen system for the pilot and/or co-pilot

List of Current Configurations:

Configuration: C510-G1- This configuration represents a twin engine turbofan, land, retractable gear, low wing, glass-cockpit airplane representative of a Cessna Citation Mustang, equipped with a three-screen G1000 avionics package.

Section 2: Description of Hardware and Software Components

All Configurations Include:

Qty	Type	Manufacturer	Name	Description/Function	Configuration
1	Software	Microsoft	Windows 7	Operating system.	All
1	Software	Lockheed Martin	Prepar3D	Simulation engine	All
1	Software	Redbird Flight Simulations	RB Sim	Simulation control and component integration.	All
1	Software	Redbird Flight Simulations	FMX Control	Motion system control and component integration. (depending on serial number)	All
1	Software	Redbird Flight Simulations	Instructors Station	Environmental, location and failure controls with map, track and glideslope display.	All
1	Software	Redbird Flight Simulations	Parrot and Cygnus	Optional software providing ATC and location services	All
1	Software	Mindstar Aviation	Redbird G1000 (C510) and C510 Gauge	Virtual flight instruments, radios, gauges, indicators, alerts, misc. instruments and logic controls for simulated systems.	All
1	Hardware	Redbird Flight Simulations	Simulation Computer	Host computer for flight simulation engine, simulation control software, airplane systems and instruments.	All
1	Hardware	Redbird Flight Simulations	Motion Control Computer	Host computer for FMX Control. (depending on serial number)	All
1	Hardware	Industry Standard	Instructors Station Computer	Host computer for Instructors Station.	All
1	Hardware	Redbird Flight Simulations	Motion Platform	Gimbaled, steel motion platform with movement in pitch, roll and yaw. Includes all motors, sensors and safety controls. (depending on serial number)	All
1	Hardware	Redbird Flight Simulations	Cockpit Enclosure	Cockpit enclosure to exclude distractions.	All
6	Hardware	22” Industry Standard	LCD	Flat Panel displays for exterior views.	All
3	Hardware	19” Industry Standard	LCD	Flat Panel displays for virtual instruments.	All
1	Hardware	Redbird Flight Simulations	Dual Yoke	Dual, Control loaded pitch and roll controller with switches and buttons for airplane systems operation.	All
1	Hardware	Redbird Flight Simulations	Throttle Quadrant	Twin, turbine engine controls with throttle levers, flap position lever and pitch, elevator, aileron and rudder trim controls with indicators for each, FMS Keypad and Emergency gear extension handle.	All
1	Hardware	Redbird Flight Simulations	Dual Rudder Pedals	Dual Rudder control pedals with toe brakes.	All
1	Hardware	Redbird Flight Simulations	Switch Panel	Lower switch panel with airplane configuration and systems controls.	All
1	Hardware	Redbird Flight Simulations	Left Switch Panel	Left side switch panel with airplane systems	All
1	Hardware	Redbird Flight Simulations	Oxygen System	Optional oxygen system for the pilot and/or copilot	All
1	Hardware	Redbird Flight Simulations	Autopilot Control	Autopilot control box mounted under eyebrow	All
1	Hardware	Redbird Flight Simulations	Circuit Breaker Panel	Pilot side circuit breaker panel.	All
1	Hardware	Redbird Flight Simulations	Circuit Breaker Panel	Co-pilot side circuit breaker panel.	All
1	Hardware	Redbird Flight	Instrument	Flight instruments, radios, airplane	All

		Simulations	Controls	configuration and systems controls.	

Statements of Compatibility of Software and Hardware AC61-136, Appendix 2, 4. b.:

The device and all configurations included in this Qualification and Approval Guide (QAG) meet the requirements for the Compatibility of Software and Hardware.

Microsoft Windows 7 (Operating System)

“This is to certify that Microsoft Corporation, the owner and developer of the Windows 7 operating system, has evaluated that their operating system works with industry standard PC’s and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software.”

Lockheed Martin Prepar3D (Simulation Engine)

This is to certify that Lockheed Martin, the owner and developer of the Prepar3D Simulator Engine, has evaluated that their software application works with industry standard PC’s and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software.

Mindstar Aviation Redbird G1000 (C510) and C510 Gauge "This is to certify that Mindstar Aviation, the owner and developer of the Redbird G1000 (C510) and C510 Gauge, and Redbird Flight Simulations, the hardware manufacture and component integrator, have demonstrated that the Mindstar Aviation software package is fully compatible with the Redbird Flight Simulations C510 and all configurations of that model. Mindstar Aviation and Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations."

Section 3: Statements of Compliance

All Configurations, as noted, meet AC 61-136, Appendix 2, Paragraph 8 BATD Design Criteria:

a. Controls:

(1) Meets Requirements: Physical flight and airplane system controls are recognizable as to their function and how they are manipulated solely from their appearance. There is no keyboard, mouse, or gaming joystick inside the Redbird C510 Cockpit. See Figures: 2.1-2.9, 3.1, 4.1, 5.1-5.10 and 6.2 - 6.4

(2) Meets Requirements: Virtual controls are limited to the initialization of the device and instructor input. Avionics are virtually displayed but are operated with physical buttons and knobs that closely represent the airplane. See Figures: 2.1-2.9, 3.1, 4.1, 5.1-5.10 and 6.2 - 6.4

(3) Meets Requirements: No keyboard or mouse is used to set or position any feature of the device. The required and additional equipment is operated in the same manner in which it would be operated in the represent airplane.

(4) Meets Requirements: The arrangement, appearance, and operation of controls, instruments, and switches closely represent the airplane, including: battery, alternators, fuel boost pumps, avionics master, pitot heat, and required lights. Magnetos are not applicable to the airplane represented. See Figures: 2.1-2.9, 3.1, 4.1, 5.1-5.10 and 6.2 - 6.4

(5) Meets Requirements: During start-up of the device the PC configuration and file inventory is examined to ensure only tested software is installed. In the event additional files are found, an initialization error will be displayed and logging of the flight training data to the student key will not be allowed.

b. Control Requirements:

(1)(a): Meets Requirements: Dual self-centering displacement yokes that are interlinked and allow for continuous adjustment of pitch and bank. The yoke handles are airplane style with electric trim, autopilot controls, a push to talk switch, a transponder ident button and other controls as applicable to the airplane. See Figures: 2.1 - 2.9

(1)(b): Meets Requirements: Dual self-centering rudder pedals with toe brakes that are interlinked and allow for continuous adjustment of yaw. See Figure: 4.1

(1)(c): Meets Requirements: Power control lever that allows for continuous movement from idle to max power, with beta and reverse settings as applicable to the airplane. See Figure: 3.1

(1)(d): Meets Requirements: Power control levers as applicable to the airplane. See Figure: 3.1

(1)(e): Meets Requirements: Controls as applicable to the airplane represented including:

- Wing flaps
- Pitch trim
- Clock and timer
- Gear handle
- Transponder
- Altimeter
- Microphone with push-to-talk switch
- Carburetor heat is not included as it is not applicable
- Cowl flaps are not included as they are not applicable

See Figures: 5.1 - 5.10 and 6.2 – 6.4

c. Control Inputs:

(1) Meets Requirements: The transport delay for all control inputs is 300 milliseconds or less. The required transport delay is tested at system start-up by internal start/stop timers. Redbird Flight Simulations is the manufacturer of all control inputs and uses USB 2.0 which exceeds this requirement. Redbird Flight Simulations certifies that the C510 meets this standard.

(2) Meets Requirements: Each control input is tested at training session start-up. If any device fails to meet the requirements of Appendix 2, Paragraph 8, Section c, item (1) a warning message is displayed and the device will not start a training session.

d. Display Requirements:

(1) Meets Requirements: See individual configuration.

(2) Meets Requirements: All instrument displays included in each individual configuration are visible during all flight operations. The displays are updated at rate that insures that the image of the instrument does not appear to: be out of focus, illegible, jump, step, jagged, nor is it distracting.

(3) Meets Requirements: Display updates exceed 60Hz. All replicated instruments display changes equal or greater than the values required by this item.

(4) Meets Requirements: Displays reflect the dynamic behavior of an actual airplane in flight.

e. Flight Dynamics Requirements:

(1) Meets Requirements: All flight dynamics are comparable to the way the represented airplane performs and handles.

(2) Meets Requirements: All simulated airplane performance parameters are comparable to the airplane represented.

(3) Meets Requirements: Vertical lift component changes as a function of bank, comparable to the airplane represented.

(4) Meets Requirements: Changes in flap setting and gear position are accompanied by changes in flight dynamics, comparable to the airplane represented.

(5) Meets Requirements: The handling and performance qualities of the simulated airplane are affected by the presence and intensity of wind and turbulence in manner comparable to the airplane represented.

f. Instructional Management Requirements:

(1) Meets Requirements: The instructor has the ability to pause the system at any time using either the buttons on the instructor's console or the buttons on the instrument panel.

(2) Meets Requirements: The instructor has the ability to manipulate the parameters listed in this item independently of the simulation.

(3) Meets Requirements: The system records both the horizontal and vertical track of the airplane movement and this can be view in real-time or at a later time.

(4) Meets Requirements: The instructor is able to disable instruments prior to or during a training session and simulate failure of the instruments without stopping or freezing the simulation to affect the failure.

(5) Meets Requirements: The device includes a navigational data base that covers all of North America and is based on procedures as published per 14 CFR part 97.

All configurations, as noted meet AC 61-136, Appendix 3, Paragraph 3 AATD Design Criteria:

a. AATD Design Criteria – All Configurations

(1) Meets Requirements: The device enclosure and cockpit design and instrument panel arrangement represent a Cessna Citation Mustang.
See Figures: 1.1 and 6.2 – 6.4

(2) Meets Requirements: Cockpit knobs/system controls/switches/switch panels are realistic in size and design appropriate to each intended function, in the proper position and distance from the pilot's seated position, and representative airplane. See individual configuration. See Figures: 2.1-2.9, 3.1, 4.1, 5.1-5.10 and 6.2 - 6.4

(3) Meets Requirements: Primary flight and navigation instruments are life size and exhibit neither stepping nor excessive transport delay. They are arranged so as to observe trends and provide a realistic scan pattern. See individual configuration. See Figures: 1.1 and 6.2 – 6.4

(4) Meets Requirements: See individual configuration.

(5) Meets Requirements: See individual configuration.

(6) Meets Requirements: See individual configuration.

(7) Meets Requirements: Manual and electric pitch trim that move an electric trim indicator in a ratio representative of the airplane. Electric pitch trim switches are located on both yokes and when activated move the manual trim well as appropriate to the airplane represented. See Figures: 2.2, 2.9, 3.1

(8) Meets Requirements: All visual cues for both day and night, VFR and IFR are provided. See Figure: 6.2

(9) Meets Requirements: Instructors station is run from a "tablet" style or laptop PC from inside or outside the device. The instructor can oversee the pilot's horizontal and vertical flight profiles in real time and space. The instructor has the ability to complete all functions listed in this item without interrupting the flight. See Figure: 6.1

(10) Cockpit seats are adjustable fore and aft. They also swivel to allow easy ingress and egress.

Configuration: C510-G1

This configuration represents a twin engine turbofan, land, retractable gear, low wing, airplane representative of a Cessna Citation Mustang

Meets all requirements under "All Configurations"

AC 61-136, Appendix 2, Paragraph 8: BATD Design Criteria

d. Display Requirements:

(1) This configuration has instruments and indicators appropriate and applicable to the airplane being represented:

Electronic Redbird G1000 (C510) with dual PFDs and one MFD with reversionary mode and back-up flight and navigation instruments. See Figure: 1.1
Sensitive altimeter that meets the requirements listed in this item.
(Dual Redbird G1000 (C510) PFDs and back up altimeter)
Magnetic compass.
Heading indicator that meets the requirements listed in this item.
(Dual Redbird G1000 (C510) PFDs)
Airspeed indicator that meets the requirements listed in this item.
(Dual Redbird G1000 (C510) PFDs and back up airspeed indicator)
Vertical speed indicator that meets the requirements listed in this item.
(Dual Redbird G1000 (C510) PFDs)
Gyroscopic rate-of-turn indicator that meets the requirements listed in this item.
(Dual Redbird G1000 (C510) PFDs)
Slip and skid indicator that meets the requirements listed in this item.
(Dual Redbird G1000 (C510) PFDs)
Engine instruments applicable to the airplane being represented.
(Redbird G1000 (C510) dual PFDs and one MFD)
Suction gauge applicable to the airplane being represented.
(Redbird G1000 (C510) MFD)

Flap setting indicator applicable to the airplane being represented.
Pitch trim indicator that meets the requirements listed in this item.
Communication radio that meets the requirements listed in this item. (Redbird G1000 (C510))
Navigation radio that meets the requirements listed in this item. (Redbird G1000 (C510))
Transponder panel that meets the requirements listed in this item. (Redbird G1000 (C510))
Fuel quantity indicators that meet the requirements listed in this item. (Redbird G1000 (C510) dual PFDs and one MFD)

AC 61-136, Appendix 3, Paragraph 3: AATD Design Criteria

a. AATD Design Criteria – Configuration: C510-G1

(2) Appropriate cockpit knobs/ system controls/switches/switch panels in realistic size and design to each intended function, in the proper position and distance from the pilot's seated position that are representative of the airplane simulated.
See Figures: 2.1-2.9, 3.1, 4.1, 5.1-5.10 and 6.2 - 6.4

(3) Life sized Redbird G1000 (C510) with dual PFDs and one MFD with dual communication and navigation radios, autopilot, audio panel, transponder, and back up heading indicator, attitude indicator and altimeter representative of airplane being simulated. Instruments exhibit neither stepping nor excessive transport delay, and are arranged so as to observe trends and provide a realistic scan pattern.
See Figure: 1.1

(4) Redbird G1000 (C510) with dual PFDs and one MFD with dual communication and navigation radios
See Figure: 1.1

(5) Redbird G1000 (C510). See Figure:1.1

(6) Redbird G1000 (C510) autopilot and flight director displayed on dual Redbird G1000 (C510) PFDs as appropriate to represented airplane. See Figure: 1.1

Section 4: Figures and Photos

Group 1: Virtual Instrument Controls Overlay

Figure 1.1: C510-C1 Instruments



Group 2: Yoke

Figure 2.1: Pilot Yoke



Figure 2.2: Pilot Yoke Switches Left Side



Figure 2.3: Pilot Yoke Switches Left Side

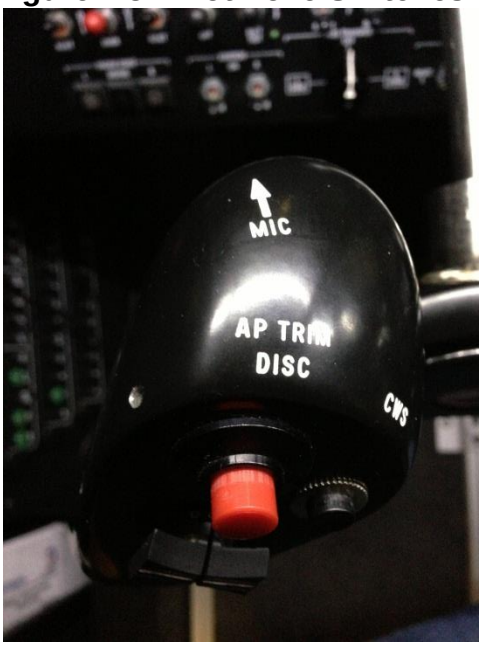


Figure 2.4: Pilot Yoke Switches Right Side



Figure 2.5: Pilot Yoke Switches Right Side



Figure 2.6: Co-Pilot Yoke

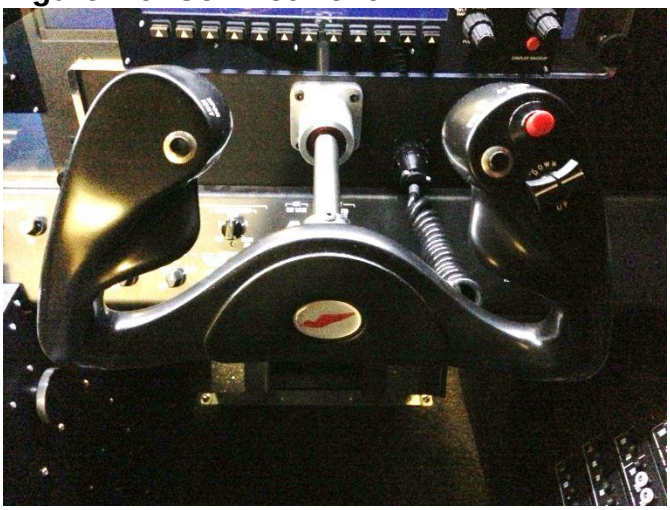


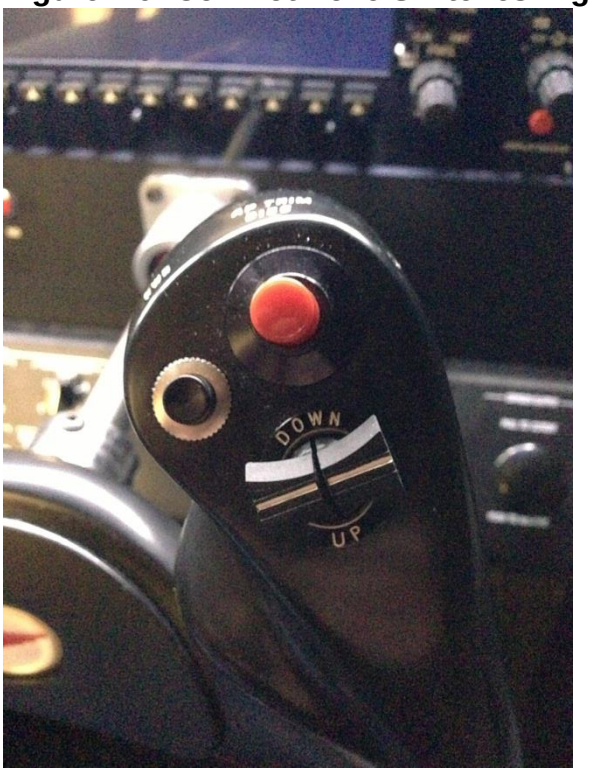
Figure 2.7: Co-Pilot Yoke Switches Left Side



Figure 2.8: Co-Pilot Yoke Switches Left Side



Figure 2.9: Co-Pilot Yoke Switches Right Side



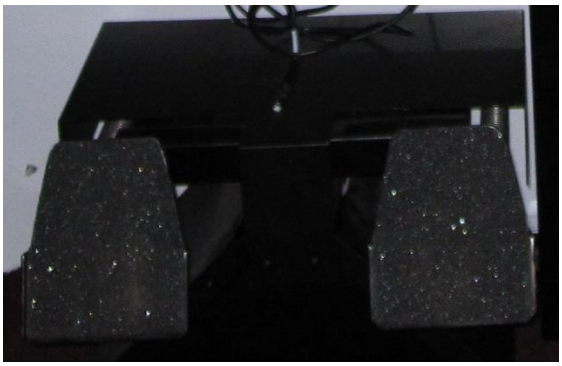
Group 3: Throttle Quadrant

Figure 3.1: Throttle Quadrant



Group 4: Rudder Pedals

Figure 4.1: Rudder Pedals Pilot and Co-Pilot



Group 5: Switches and Other Controls

Figure 5.1: FMS Keypad

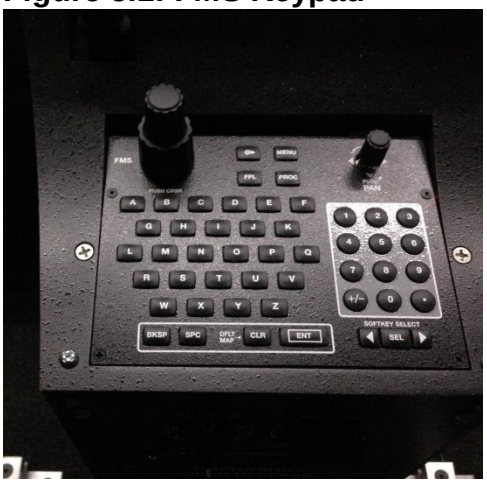


Figure 5.2: Circuit Breaker Panel Pilots Side



Figure 5.3: Circuit Breaker Panel Co-Pilots Side

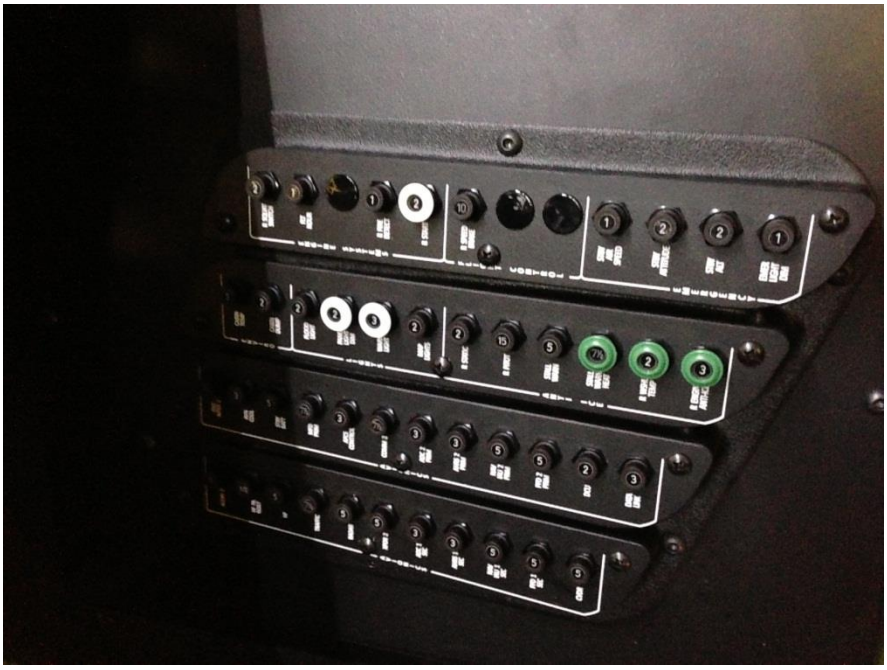


Figure 5.4: Autopilot Control Panel



Figure 5.5: Lower Switch Panel Pilots Side 1



Figure 5.6: Lower Switch Panel Pilots Side 2



Figure 5.7: Lower Switch Panel Center



Figure 5.8: Lower Switch Panel Co-Pilots Side



Figure 5.9: Gear Handle

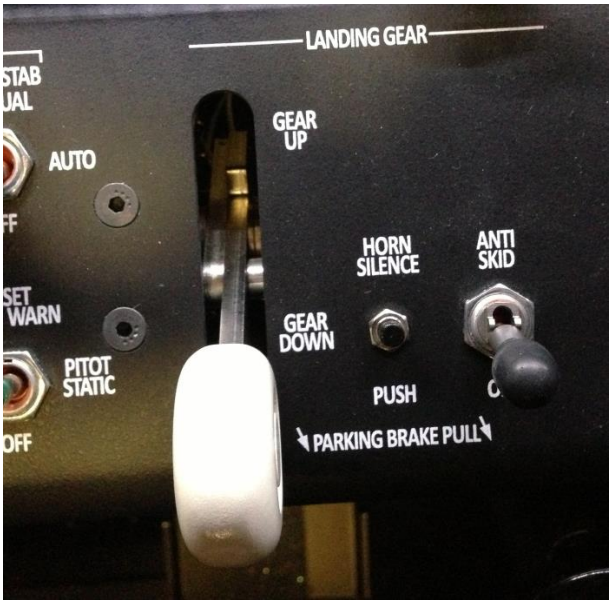


Figure 5.10: Emergency Gear Extension Handle



Group 6: Other Images

Figure 6.1: Instructors Station Tablet PC



Figure 6.2: Assembled Cockpit



Figure 6.3: Assembled Cockpit



Figure 6.4: Assembled Cockpit



Figure 6.5: Motion Platform and Cockpit Enclosure



Appendix 1: Advanced ATD Checklist

Functions and Maneuvers	Satisfactory? Yes/No
a. Preparation for Flight	
(1) Preflight	Yes
b. Pre-Takeoff	
(1) Engine start	Yes
(2) Brake operation	Yes
c. Takeoff	
(1) **AIRPLANE Takeoff**	
(i) Power plant checks	Yes
(ii) Acceleration characteristics	Yes
(iii) Nose wheel and rudder steering	Yes
(iv) Effect of crosswind	Yes
(v) Instrument	Yes
(vi) Landing gear, wing flap operation	Yes
(2) **HELICOPTER Takeoff**	
(i) Power plant checks	N/A
(ii) From Hover	N/A
(iii) From Ground	N/A
(iv) Vertical	N/A
(v) Running	N/A
d. In-flight Operation	
(1) **AIRPLANE In-flight Operation**	
(i) Climb	Yes
(a) Normal	Yes
(b) One engine inoperative procedures	Yes
(ii) Cruise	
(a) Performance characteristics (speed vs. power)	Yes
(b) Normal and steep turns	Yes
(c) Performance turns	Yes
(d) Approach to stalls, i.e., stall warning (cruise, takeoff/approach, and landing configuration)	Yes
(e) High angle of attack maneuvers (cruise, takeoff/approach, and landing)	Yes
(f) In-flight engine shutdown	Yes
(g) In-flight engine restart	Yes
(iii) Descent	Yes
(a) Normal	Yes
(b) Maximum rate	Yes
(2) **HELICOPTER In-flight Operation**	
(i) Hovering	N/A
(a) Forward	N/A
(b) Rearward	N/A
(c) Sideward	N/A

(d) Turns	N/A
(ii) Climb	N/A
(iii) Cruise	N/A
(a) Performance characteristics	N/A
(b) Turns	N/A
i. Recovery	N/A
ii. Skidding	N/A
iii. Slipping	N/A
(c) In-flight engine shutdown	N/A
(d) In-flight engine restart	N/A
(iv) Descent	N/A
e. Approaches	
(1) Non-precision	Yes
(i) GPS-WAAS	Yes
(ii) GPS-LAAS	Yes
(iii) All engines operating	Yes
(iv) One or more engines inoperative	Yes
(v) Approach procedures (NDB, VOR, DME Arc, LOC/BC, LOC, LDA, SDF, ASR, LNAV/VNAV, GPS, & LPV, and additional types when developed and certified)	Yes
(vi) Missed approach (all engines operational, one or more engines inoperative)	Yes
(2) Precision	Yes
(i) PAR – normal	Yes
(ii) ILS	Yes
(iii) GLS	Yes
(iv) Effects of crosswind	Yes
(v) With engine(s) inoperative	Yes
(vi) Missed approach	Yes
(a) Normal	Yes
(b) With engine(s) inoperative	Yes
(c) From steep glide slope	Yes
f. Surface Operations (AIRPLANE-Post Landing)	
(1) Landing roll	Yes
(2) Braking operation	Yes
(3) Reverse thrust operation, if applicable	N/A
g. Any Flight Phase	
(1) Aircraft and power plant systems	Yes
(i) Electrical	Yes
(ii) Flaps (**AIRPLANE**)	Yes
(iii) Fuel and oil	Yes
(iv) Landing gear (**AIRPLANE**)	Yes
(2) Flight management and guidance systems	Yes
(i) Automatic pilot	Yes
(ii) Flight director/system displays	Yes
(iii) Navigation systems	Yes
(iv) Stall warning/avoidance (**AIRPLANE**)	Yes

(v) Multi function displays	Yes
(vi) Primary flight displays	Yes
(3) Airborne procedures	Yes
(i) Holding	Yes
(4) Engine shutdown and parking	Yes
(i) Systems operation	Yes
(ii) Parking brake operation (**AIRPLANE**)	Yes